Flowchart

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A flow chart is a graphical representation of an algorithm or process. It uses standardized symbols connected by arrows to show the step by step flow of control in solving a problem.

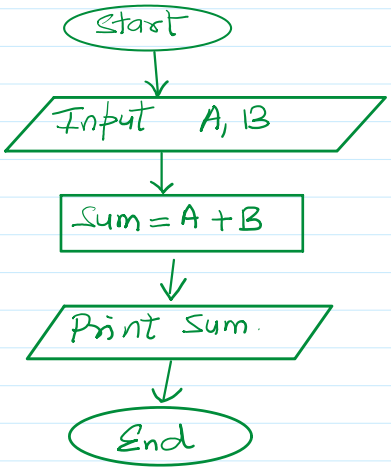
Definition: A flow chart is a diagram that represents a sequence of instructions using symbols, and flow lines.

It is widely used in:

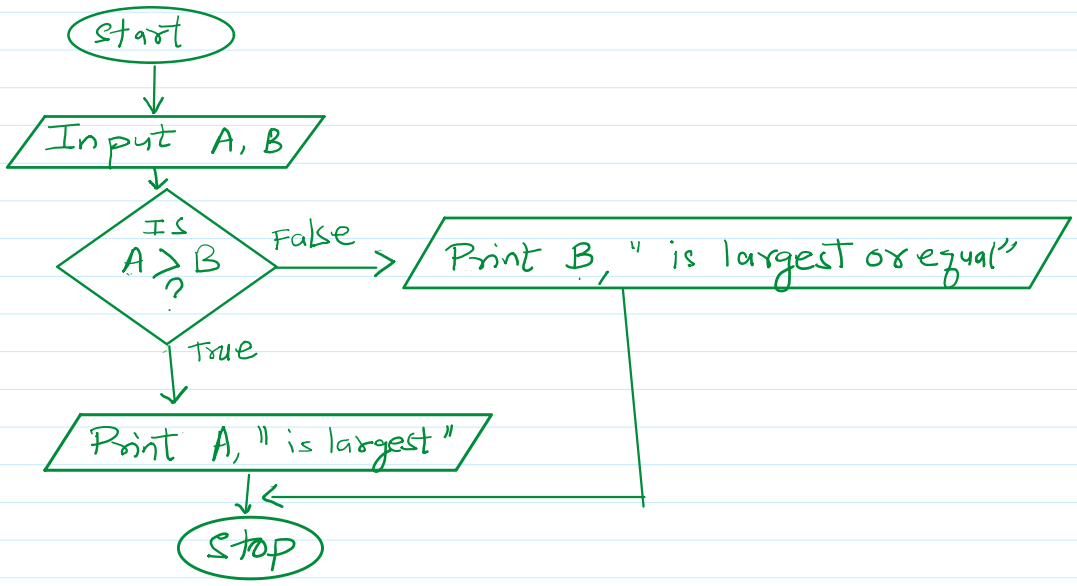
- Programming(before coding)
- System design
- Problem solving

Symbol	Name	Purpose
oval	Start / End	Beginning or termination.
rectangle	Process	Calculation / instruction.
Parallelogram	Input / Output	Read or display
Diamond	Decision	Condition checking
→	Flow line.	Direction of flow.
C	Connector	Used to connect with another flowchart.

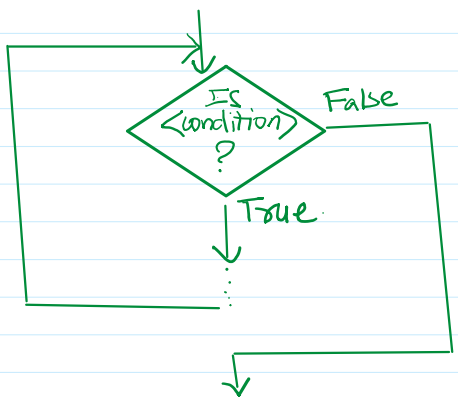
Example-1 Add two numbers and print the sum.



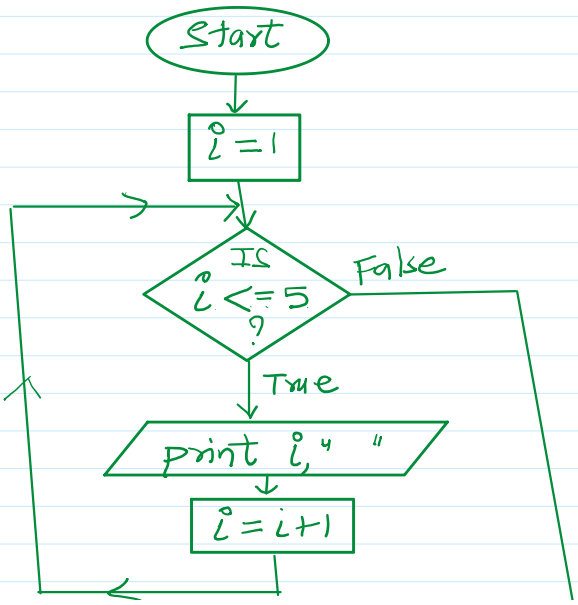
Example: Largest of two numbers.



Loop or Iteration or Repetition.

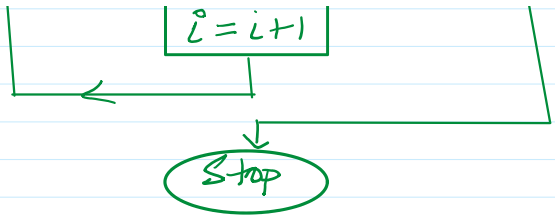


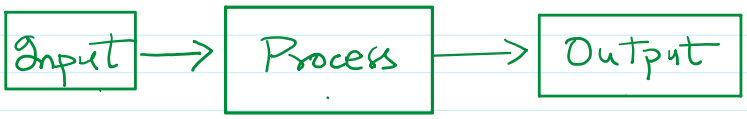
Example: Print the numbers from 1 to 5.



Output:

1_2_3_4_5_





Functions :

$$f(x) = \boxed{x^3 + 3x^2 + 3x + 1}$$ [x can be any real number]

parameter (input) — process — output.

$$f(2) = 2^3 + 3 \cdot 2^2 + 3 \cdot 2 + 1 = 8 + 12 + 6 + 1$$

$$= 27 \leftarrow \text{output}$$

what the domain of $f(x)$?

Any real number.

Complex Problems how to solve a complex problem?

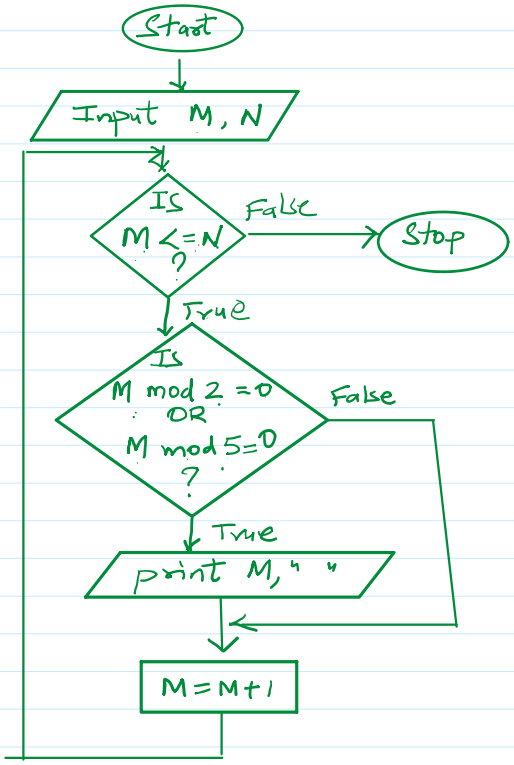
Analyze the problem.
Divide the problem in multiple functions.
collect the output of all functions,
And, find the final solutions

Subroutine ← It is used to solve a part of complex problem.

Draw a flow chart to print all the numbers either divisible by 2 or 5, from M to N; [where $M <= N$].

For example: M = 20
N = 50

Output: 20, 22, 24, 25, 26, 28, 30, 32, 34, 35, 36, 38, 40, 42, 44, 45, 46, 48, 50.



M: ~~20~~, ~~21~~, ~~22~~, 24. N: 50

20
22

mod ← mean remainder in coding

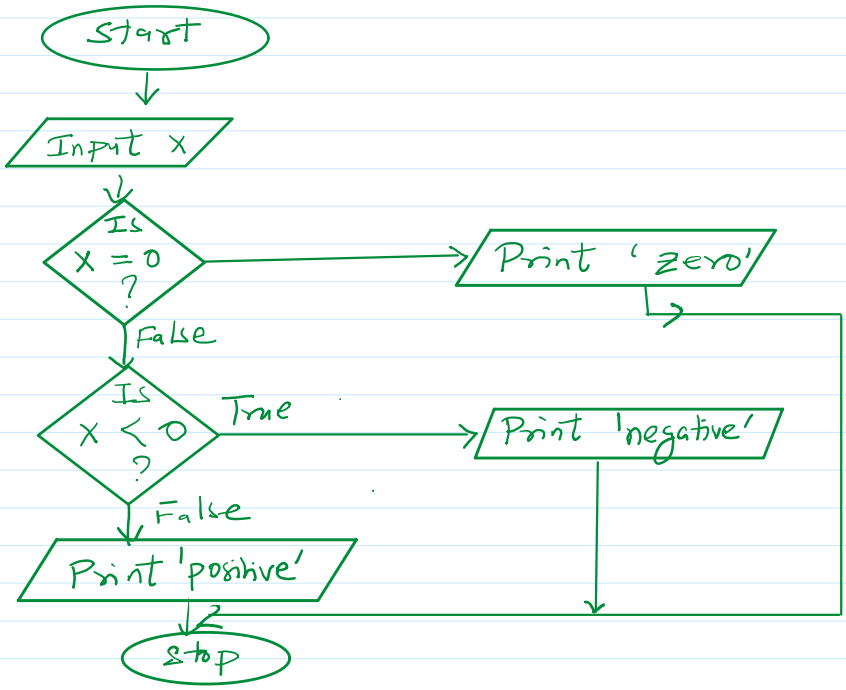
Advantages of Flowcharts

* Easy to understand logic visually
* Helps in debugging before coding
* Improves communication between developers
* Useful for beginners.

Disadvantages

* Time-consuming to draw.
* Difficult for complex systems.
* Hard to modify once drawn.

$$\frac{C}{5} = \frac{F-32}{9} \Rightarrow \boxed{F = \frac{9 \cdot C}{5} + 32}$$

1. Draw a flow chart to convert temperature from Celcius to Fahrenheit.
2. Draw a flowchart of calculate the area of rectangle.
3. Draw a flowchart to check whether a number is positive, negative, or zero.

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